

Marc Lally

Senior Software Engineer | Node | React | Serverless | AWS

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TLDR;

- Senior Software Engineer | TypeScript · Node · React · Python · Rust | PostgreSQL · MongoDB · GraphQL | AI Workflows & Agents | PyDantic AI · LangSmith AI · Agents SDK · Gradio AI | DevOps | Terraform | Serverless | Docker | Likes Challenges

PROFILE

- Senior Software Engineer with extensive experience translating complex business models into robust, scalable technical solutions. Design and evolve distributed systems optimizing for reliability, cost efficiency, and velocity. Lead service-oriented and serverless architecture with clear boundaries, data ownership models, and operational visibility. Establish API standards (REST/GraphQL) with rigorous versioning and governance. Drive architectural direction through data-driven strategy, mentoring senior engineers and bridging engineering, product, and business stakeholders.

EDUCATION

B.Sc. (Hons) - 2:1 in Computer Science

Sept 2018 – June 2021

University College London

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate Level - Validation of cloud architecture expertise
- AWS Certified Engineer - Associate Level - Validation of cloud development expertise

EXPERIENCE

CTO & Founder, PayStacks

Feb 2026 – present

- Full-stack development of enterprise-grade AI, highly secure, Payment Services Gateway API in Rust, Website and frontend in Next.js, React and various frontend javascript frameworks and libraries to support extension of functions or new features.
- Architect of a payment processing service API in Rust, using HECE encryption/decryption. Built a clean RESTful API with SDKs across every major language, runtime, and mobile platform. Native iOS and Android SDKs with pre-built UI components, plug-and-play CMS integrations, and a full sandbox for testing without limits.
- Designed and built a Hybrid AI ML model which scores every transaction in real time. Review queue, threshold controls, and online feedback learning to considerably reduce fraud with a comprehensive scoring system that evaluates risk across multiple dimensions.
- Hydra Ephemeral Channel Encryption derives a per-request AES-256-GCM key via HKDF-SHA256, bound with HMAC-SHA256. Every API call gets a unique key tied to the exact request path and timestamp. Highly secure transaction and data encryption protocols.
- Built and deployed many Agents which handle social integrations, posts, customer support, sales Agents, security Agents and Agents which mock the transactional process to improve and develop user integrations. Integrated modern features, libraries and workflows into the frontend with extensive Testing using viTest and Jest with Playwright for E2E testing to ensure the quality of the product and the user experience.
- Developed agentic workflows for automated technical design reviews, leveraging LLMs for context-aware code reasoning and architectural consistency checks. Built AI Agents with Pydantic AI, Agents SDK, LangChain, LangSmith, Gradio, FastAPI and other tools.
- Designed and built modern UI interface with animated dashboards, workbench and other tools using Next.js, React and various frontend javascript frameworks featuring extensive audit and security logs, with considerable encryption.

Senior Fullstack Engineer, Cyber & Specialist Ops, Virgin Atlantic

Feb 2025 – Jan 2026

- Architected AI-powered security workflows using vector embeddings to surface anomalies in high-throughput identity logs, improving threat detection speed.
- Delivered high-throughput Node.js microservices on AWS Lambda and DynamoDB, implementing fine-grained IAM policies for secure deployment. Optimized AWS Concurrency and execution context reuse to maintain good latency for JWT-authenticated traffic volumes.
- TDD with Jest and Supertest, raising code coverage and cutting production regressions. Established contract testing with Pact to stabilize distributed service boundaries.
- Hardened security posture by integrating threat modeling and automated dependency scanning into CI/CD pipelines. Built secure-by-default patterns for OAuth 2.0 and JWT identity flows.
- Aggressive profiling with AWS X-Ray and performance tuning of DynamoDB GSI/LSI patterns. Scaled platform throughput via intelligent request batching and SQS-driven async processing.
- Implemented comprehensive observability with structured logging, distributed tracing, and custom metrics enabling rapid incident resolution (New Relic). Built backend functionality in Node/AWS and integrated internal tooling into the workflows.
- Built AI Agents using Pydantic AI, Agents SDK, LangChain, LangSmith, Gradio, and other tools to optimize and control the workflows and run DevOps tasks. Built Lambda functions and state machine language workflows using SML as logic multipliers.

Senior Full Stack Engineer | Tech Lead, Codeium

Mar 2023 – Dec 2024

- Full-stack development of enterprise-grade AI coding assistants, optimizing real-time inference delivery and high-throughput vector indexing workflows.
- Architected scalable React/TypeScript interfaces for complex Engineer workflows, including VS Code and JetBrains IDE integrations.
- Designed and implemented zero-trust authentication and authorization patterns for multi-tenant enterprise deployments.
- Optimized frontend performance and state management (Redux/Zustand), reducing client-side latency for large repositories.
- Established rigorous E2E testing patterns (Playwright/Cypress/Jest) and CI/CD automation, ensuring stability across diverse deployment environments.
- Developed agentic workflows for automated technical design reviews, leveraging LLMs for context-aware code reasoning and architectural consistency checks. Built AI Agents with Pydantic AI, Agents SDK, LangChain, LangSmith, Gradio, FastAPI and other tools.
- Design and build scalable frontend interfaces using Next.js, React and various frontend javascript frameworks and libraries to support extension of functions or new features.

Senior Full Stack Engineer, Shell Global Energy

July 2022 – Feb 2023

- Engineered a horizontally-scalable GraphQL layer (Apollo Server) on Node.js/Lambda with DataLoader batching and Redis-based caching strategies.
- Developed robust payment pipelines with Stripe and Adyen, implementing idempotency keys and webhook retry policies.
- Implemented fault-tolerant patterns including circuit breakers, bulkheading, and dead-letter queues using AWS Lambda, SQS, SNS and S3.
- Implemented comprehensive monitoring with CloudWatch, X-Ray, and custom dashboards for rapid incident resolution.
- Mentored 6+ engineers on distributed systems and payment processing; conducted 40+ technical interviews.
- Improved deployment velocity enabling 5+ releases daily through CI/CD automation and disaster recovery drills with Terraform and CloudFormation.

Senior Full Stack Engineer | Tech Lead, Studio Rotate

June 2021 – July 2022

- Directed technical delivery for high-visibility Next.js/Vue.js applications. Defined clean API contracts and schema governance to support extensive user sessions.
- Built and managed infrastructure using Terraform and AWS, deploying Node.js/Serverless backends with automated CI/CD pipelines via GitHub Actions.
- Elevated engineering maturity by introducing ESLint/Prettier standards, trunk-based development, and peer code review workflows.
- Implemented deep observability and alerting; analyzed performance metrics to identify and eliminate bottlenecks.
- Designed and built backend Serverless API to support builds taking place for Chillys and Big Green Egg. Built frontends in Next.js, Vue.js and React.js

Senior Backend Engineer, Sainsburys Digital

Dec 2020 – June 2021

- Modernized legacy Node.js/TypeScript services by refactoring to a modular architecture and patching critical vulnerabilities. Integrated Google DV360 API via a Next.js middleware layer.
- Optimized the critical rendering path; reduced React bundle sizes and slashed LCP through aggressive code-splitting and asset optimization.
- Hardened data security by implementing end-to-end encryption and field-level access controls using AWS KMS.
- Collaborated on GraphQL schema design and cross-team API contracts, leading architecture walkthroughs.
- Advanced PostgreSQL performance through strategic indexing, connection pooling (PgBouncer), and schema normalization.

Senior Fullstack Engineer | Tech Lead, KFC Europe

Dec 2019 – Nov 2020

- Orchestrated integrations between web ordering, POS (NCR/MICROS), and loyalty platforms. Established strict API boundaries and contract testing (Pact) to prevent integration drift across microservices.
- Designed multi-tenant AWS Lambda services secured via OAuth 2.0 and JWT, secured for high-availability fast-food infrastructure.
- Optimized PostgreSQL and DynamoDB schema designs to meet sub-50ms p95 latency targets. Implemented multi-layer caching (Redis/CloudFront) to achieve performance gains.
- Coordinated technical delivery across global teams, enabling a high-release-velocity environment (50+ monthly features) with a heavy focus on stability.
- Mentored 6 engineers through the complexities of distributed systems and payment processing.
- Aggressively tuned Lambda resource allocation and cold starts, cutting execution time and infrastructure burn. Built AWS Lambda infrastructure using Terraform and CloudFormation with SQS, SNS for high availability.

Senior Frontend Engineer, Preqin

Feb 2019 – Nov 2019

- Directed frontend architecture for a high-traffic financial platform. Combined React, TypeScript, and Elasticsearch to deliver high-performance search and data visualization for institutional investors.
- Reduced bundle sizes and significantly improved Core Web Vitals through custom Webpack/Vite configurations and code-splitting strategies.
- Implemented exhaustive test suites using Jest and React Testing Library, achieving high coverage across critical financial data flows.
- Optimized Elasticsearch query patterns and mapping, reducing search latency across massive financial taxonomies.

Senior Front End Engineer, MelodyVR

Jan 2018 – Dec 2018

- Built a custom React-based CMS on Node.js/Serverless, enabling content teams to orchestrate and distribute VR assets across cross-platform channels with zero downtime.
- Designed a scalable, component-driven UI architecture using Storybook and design tokens, which accelerated iterative feature delivery.
- Architected CI/CD pipelines using GitHub Actions and Terraform (IaC), slashing deployment cycles from hours to minutes and enabling 16x faster shipping frequency.
- Optimized asset delivery and page performance, cutting load times through aggressive code-splitting and caching strategies.

Senior Back End Engineer, Big Clever Learning

Apr 2017 – Nov 2017

- Architected a decoupled microservices platform using Node.js and Koa. Defined service boundaries via OpenAPI/Swagger 3.0 to ensure strict contract enforcement.
- Managed polyglot persistence across PostgreSQL, MongoDB, and Redis; delivers high availability and sub-50ms latency for a large user base.
- Integrated Stripe for high-volume billing, maintaining PCI compliance while managing transactions with robust idempotency and error handling.
- Containerized the platform with Docker and modernized infrastructure via Terraform, significantly reducing deployment overhead.

Senior Back End Engineer, FutureBrand

Oct 2016 – Mar 2017

- Spearheaded the integration of OAuth 2.0 and SAML 2.0, enabling secure SSO across diverse enterprise ecosystems with robust token management and key rotation in PHP.
- Developed 20+ custom plugins for Craft CMS using PHP and MySQL, delivering bespoke workflows and complex data models for global enterprise clients.
- Executed incremental refactors to modernize legacy systems while ensuring zero downtime and maintaining strict backwards compatibility.
- Optimized MySQL for CMS platform serving numerous sites; reduced page load time.

Senior Full Stack Engineer | Team Lead, Purple Bridge Publishing

Jan 2015 – Sept 2016

- Led delivery of complex enterprise web applications on LAMP stack; established engineering standards including code style guides and security policies.
- Designed normalized data models and RESTful APIs handling complex educational content management and regulatory compliance for UK schools.
- Pioneered Docker and Jenkins CI/CD; reduced deployment time from 2 hours to 10 minutes enabling daily releases.
- Mentored engineering team of 6 through code reviews, interviews, and design sessions focused on architectural thinking.

Senior Full Stack Engineer, MediaCode London

May 2006 – Oct 2014

- Built and maintained custom enterprise software and system integrations for financial services, healthcare, and e-commerce clients delivering solutions across diverse domains.
- Owned end-to-end requirements discovery and specification; produced technical architecture diagrams and design documents aligned with business objectives.
- Systematically optimized legacy system performance; implemented database indexing, query optimization, and caching improving response times
- Provided 24/7 production support with critical issue resolution; maintained uptime SLAs and built operational runbooks.

Business Support Analyst, Orange

Mar 2001 – Feb 2006

- Engineered technical integrations and automated reporting for telecommunications services serving 10 European countries; connected disparate legacy systems enabling real-time data visibility.
- Established development standards and code review practices across operations teams.
- Spearheaded digital transformation from manual processes to automated workflows; built tools and dashboards reducing administrative overhead.
- Designed data warehouses and reporting schemas enabling business intelligence for C-suite strategic decisions.

Software Developer, HSBC

1997 – 2001

- Developed mission-critical online banking software with stringent security requirements; implemented SSL encryption, input validation, and threat modeling.
- Ensured regulatory compliance with banking standards (BSA, AML, KYC); embedded controls into product design and testing.
- Facilitated cross-functional collaboration across product, systems, QA, operations, and risk teams.
- Developed core banking features (transaction processing, account management, reporting) with rigorous QA and security testing.

Software Developer, Next Ltd

1994 – 1997

- Built customer account and transaction web applications for Next's online division; implemented transactional integrity and comprehensive audit logging.
- Contributed across full lifecycle including requirements analysis, design, implementation, testing, and deployment meeting regulatory timelines.
- Implemented security best practices including password hashing, secure session management, and encryption ensuring PCI compliance.
- Designed database schemas for transaction systems; understood ACID properties and data consistency requirements for critical systems.

SKILLS

Senior Software Engineer: Node/React | Javascript | AWS | Serverless | AI Workflows/Agents

Core Expertise: TypeScript · Node · React · Python · Rust | PostgreSQL · MongoDB · GraphQL | AI Workflows & Agents | PyDantic AI · LangSmith AI · Agents SDK · Gradio AI | DevOps | Terraform | Serverless | Docker

Distributed Systems: Scalable, high-throughput production environments

Platform Strategy: Service-oriented and serverless architecture design with operational excellence

Multi-Stack: TypeScript/Node, Python, Rust

Security & Compliance: OAuth 2.0, SAML, SSO, JWT and high-availability infrastructure